

Abeer Banerjee

PhD Researcher | Computational Imaging

Inverse Problems ◦ Machine Learning ◦ Medical Imaging

[abeerbanerjee.github.io](https://github.com/abeerbanerjee)

Munich, Germany

abeer.ceeri20a@acsir.res.in 

[LinkedIn Profile](#) 

[Google Scholar Profile](#) 



Professional Summary

- PhD researcher specializing in computational imaging and physics-informed machine learning for inverse imaging problems. Experienced in deep learning for lensless imaging, optoacoustic microscopy, and neuromorphic vision.
- Currently a DAAD PhD Fellow at the TUM Center for Translational Cancer Research (Transl-TUM), and a guest researcher at Institute of Biological and Medical Imaging (IBMI), Helmholtz Center Munich.

Research Highlights

Research Area	Computational Imaging, Scientific Machine Learning, Inverse Problems, Computer Vision, Image Processing.
Research Output	12+ peer-reviewed publications, including 6+ first-author papers in IEEE Transactions, Elsevier journals, BMVC, ICIP, IJCNN, etc., 2 registered copyrights, and 1 manuscript in preparation.
Research Experience	Five years of research experience spanning deep learning for imaging, especially lensless imaging, neuromorphic imaging and optoacoustic imaging.

Research Experience

Research Theme: Physics-informed Computational Imaging and Inverse Problems

Technical University of Munich, Germany (2024-Present)

CSIR–Central Electronics Engineering Research Institute, India (2020-2026)

- Designed physics-informed learning-based reconstruction methods integrating forward imaging models with deep neural networks for lensless imaging.
- Investigated implicit neural representations and generative priors for solving inverse imaging problems. Published research on SCI journals and international conferences.
- Developed GAN-based historical grayscale image colorization and restoration.

Research Theme: Biomedical Optoacoustic Imaging

Institute of Biological and Medical Imaging, Technical University of Munich, Germany

- Investigating physics-informed deep learning frameworks for improving axial resolution. Building simulation and reconstruction pipelines for quantitative optoacoustic imaging.
- Developing model-based reconstruction algorithms for optoacoustic microscopy using characterized system impulse responses.

Research Theme: Neuromorphic Vision

CSIR–Central Electronics Engineering Research Institute, India (2022-2024)

- Developed deep learning methods for eye-gaze estimation under extreme low-light conditions using event-based sensors.
- Contributed to face anti-spoofing, human activity recognition and privacy-preserving vision systems. Published research presented at IEEE IJCNN and other international conferences.

Selected Publications

Journal Articles

- **A. Banerjee**, and S. Singh, *Towards Physics-informed Cyclic Adversarial Multi-PSF Lensless Imaging*, IEEE Transactions on Emerging Topics in Computational Intelligence, 2025, IF: 8.0. [[Paper](#)].
- **A. Banerjee**, and S. Singh, *Towards Lensless Image Deblurring with Prior-Embedded INRs*, Engineering Applications of Artificial Intelligence, Elsevier, 2025, IF: 8.0. [[Paper](#)].
- H. Kumar*, **A. Banerjee***, et al., *ParaColorizer: Realistic Image Colorization using Parallel Generative Networks*, The Visual Computer, Springer, 2023, IF: 3.4 *Equal contribution [[Paper](#)]
- S. Prasad, N. Mehta, **A. Banerjee**, et al., *JS-SpoofNet: Jointly Supervised Parallel Branched Neural Network for Spoof Detection*, Neurocomputing, 2023, IF: 6.5. [[Paper](#)].

Selected Peer-reviewed Conference Publications

- **A. Banerjee**, et al., *Reconstructing Synthetic Lensless Images in Low-Data Regime*, BMVC 2023. h5-index: 77. [[Paper](#)]
- **A. Banerjee**, et al., *Physics-informed Deep Deblurring: Over-parameterized Vs. Under-parameterized*, IEEE ICIP 2023, h5-index: 61. [[Paper](#)]
- **A. Banerjee**, et al., *Generalized Gaze-Estimation in Low-light with Event-driven Neural Networks*, IEEE IJCNN 2024 (Oral) h5-index: 60. [[Paper](#)]
- **A. Banerjee**, et al., *Physics-Informed Neural Reconstruction in Optoacoustic Microscopy*, IEEE ISBI 2026 h5-index: 59 (Peer-reviewed Poster Presentation)

Complete publication list: [Google Scholar Profile](#)

Education

Integrated PhD in Computational Imaging 2020–Present

Academy of Scientific and Innovative Research (AcSIR), India [Anabin H+]

Advisors: Dr. Sanjay Singh, Prof. Vasilis Ntziachristos

Master's Thesis: Generative Colorization of Grayscale Images [2022]

CGPA: 9.13/10

B.Tech. Electronics and Communication Engineering 2016–2020

University of Calcutta, India [Anabin H+]

CGPA: 8.64/10

Core Expertise

Research	Computational Imaging, Inverse Problems, Scientific Machine Learning, Image Reconstruction, Physics-informed Learning
Machine Learning	Physics-informed learning, implicit neural representations, generative models, representation learning
Programming	Python (advanced), MATLAB
Frameworks	PyTorch, TensorFlow, OpenCV, Docker, Git, TensorBoard, L ^A T _E X
Languages	English (fluent), Bengali (native), Hindi, German (basic)

Academic Service

- **Reviewer:** Neurocomputing, The Visual Computer, Expert Systems with Applications, Computers & Electrical Engineering, IET Computer Vision, IEEE IJCNN, ACML, ICVGIP and other international venues.
- **Session Chair:** IEEE International Joint Conference on Neural Networks (IJCNN), 2024

Honours & Awards

- DAAD Bi-national Doctoral Fellowship, Technical University of Munich (2024–2026)
- Junior and Senior Research Fellowship (GATE), Government of India (2020–2025)
- *Extracurricular:* Fine Arts (Painting) 2013: 5th-year degree with distinction, Bangiya Sahitya Parishad, RBU.